

GeoAI Urban Expansion Analysis Report (2021–2026)

This report presents a GeoAI-based analysis of urban expansion using Sentinel-2 imagery for 2021 and 2026. Random Forest classification, change detection, and deep learning techniques were used to quantify land-cover changes.

Phase 2 – Random Forest Classification

Land-cover maps were generated using Sentinel-2 bands (B02, B03, B04, B08, B11) and spectral indices including NDVI, NDWI, and NDBI.

Phase 3 – Change Detection

The classified maps were compared to identify urban growth, vegetation loss, and water changes between 2021 and 2026.

Phase 4 – Deep Learning Change Detection

A CNN model was trained on image patches and achieved approximately 94.11% training accuracy and 94.83% validation accuracy.

Phase 5 – Urban Expansion Analysis

Results

Built-up Area 2021: 162.9894 km²

Built-up Area 2026: 165.9710 km²

Net Built-up Change: +2.9816 km²

Urban Growth Area: 24.4992 km²

Urban Loss Area: 21.5176 km²

Vegetation Loss Area: 3.7849 km²

Water Change Area: 18.8686 km²

Built-up Growth Percentage: 1.83%

Conclusion

The analysis demonstrates measurable urban expansion between 2021 and 2026. The workflow successfully integrated machine learning, deep learning, and geospatial techniques to identify urban growth, vegetation loss, and water-body changes.